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Technical Audit: pmsimstats Project Family

Comprehensive Review of Three Sibling Repositories for Precision Medicine Trial Simulation

Author: Technical Audit (Claude) Date: 2026-03-18



1. Structural Inventory

1.1 Repository Overview

Attribute	orig	main	simple
Package name	pmsimstats	pmsimstats2025	pmsimstatssimple
Version	0.0.0.900	0.1.0	0.0.0.9000
License	GPL 2.0	GPL-3	GPL-3
Total R lines	3,142	15,584	562
R files (src)	11	44	1
Test files	0	5	1
Doc files	4 (README + 3 Rmd)	40+ (md, tex, Rmd)	4 (md)
Docker	No	Yes	Yes
renv	No	Yes	Yes
Makefile	No	Yes	Yes

1.2 Directory Trees

1.2.1 orig (pmsimstats)

```
pmsimstats/
+-- DESCRIPTION
+-- NAMESPACE
+-- README.md
+-- R/
|   +-- buildtrialdesign.R      (119 lines)
|   +-- censordata.R          ( 89 lines)
|   +-- comb.R                (1571 lines)
|   +-- datadocumentation.R   ( 61 lines)
|   +-- generateData.R        (223 lines)
|   +-- generateSimulatedResults.R (255 lines)
|   +-- lme_analysis.R        (217 lines)
|   +-- packagedocumentation.R ( 19 lines)
|   +-- plottingfunctions.R   (341 lines)
|   +-- scratch_WorkingScript.R (142 lines)
|   +-- utilities.R           (105 lines)
+-- vignettes/
    +-- Produce_Publication_Results_1_generate_data.Rmd
    +-- Produce_Publication_Results_2_Look_at_data.Rmd
    +-- Produce_Publication_Results_3_apply_to_actual_clinical_trial_data.Rmd
```

1.2.2 main (pmsimstats2025)

```
pmsimstats2025/
+-- DESCRIPTION
+-- NAMESPACE
+-- CLAUDE.md
+-- Dockerfile
+-- Makefile
+-- renv.lock
+-- R/
|   +-- utils.R
```

```

+-- analysis/
| +-- scripts/
| | +-- pm_functions.R          (1301 lines)
| | +-- simulation_clustered.R  (1233 lines)
| | +-- simulationplus.R        ( 995 lines)
| | +-- simulation_evenly_spaced.R ( 649 lines)
| | +-- simulation_carryover_robustness.R
| | +-- simulation_carryover_spectrum.R
| | +-- simulation_censoring.R
| | +-- simulation_clustered_exponential.R
| | +-- simulation_clustered_flexible_correlation.R
| | +-- simulation_clustered_twoweek.R
| | +-- simulation_evenly_spaced_flexible_correlation.R
| | +-- simulation_evenly_spaced_powerlaw.R
| | +-- simulation_response_params.R
| | +-- visualize_heatmaps.R
| | +-- visualize_hendrickson_style.R
| | +-- visualize_power_heatmaps_4panel.R
| | +-- plot_results.R
| | +-- plot_censoring.R
| | +-- summarize_results.R
| | +-- check_model_specification.R (x3)
| | +-- debug_effect_recovery.R
| | +-- diagnose_nonmonotonic_power.R
| | +-- example_parameter_validation.R
| | +-- profile_simulation.R
| | +-- test_*.R (5 ad-hoc test scripts)
| | +-- 00_*.R (setup scripts)
| | +-- 02_data_validation.R
| | +-- 99_reproducibility_check.R
| | +-- carryover_focus/
| |     +-- carryover_power_simulation.R
| +-- data/
| +-- output/
+-- docs/
| +-- (40+ markdown, tex, and Rmd files)
+-- tests/
| +-- testthat.R
| +-- testthat/
| | +-- test-utils.R
| +-- integration/
|     +-- test-analysis-scripts.R
|     +-- test-data-pipeline.R
|     +-- test-report-rendering.R
+-- vignettes/
    +-- 01_generate_simulation_data.Rmd
    +-- quickstart.Rmd

```

1.2.3 simple (pmsimstats-simple)

```
pmsimstats-simple/
+-- DESCRIPTION
+-- NAMESPACE
+-- CLAUDE.md
+-- Dockerfile
+-- Makefile
+-- renv.lock
+-- zzcollab.yaml
+-- .github/workflows/
|   +-- r-package.yml
+-- analysis/
|   +-- scripts/
|       +-- simulation.R          (562 lines)
|   +-- data/
|       +-- README.md
|   +-- output/
|       +-- comparison_whitepaper.md
+-- docs/
|   +-- simplification-plan.md
+-- tests/
|   +-- testthat.R
|   +-- testthat/
|       +-- test-basic.R
```

1.3 Source File Manifest

1.3.1 orig – Exported Functions (from NAMESPACE)

Function	File	Lines	Purpose
buildtrialdesign	buildtrialdesign.R	119	Construct trial design from user-friendly inputs
generateData	generateData.R	223	Core MVN data-generating process
generateSimulatedResults	generateSimulatedResults.R	255	Parameter-space iteration wrapper
lme_analysis	lme_analysis.R	217	LME model fitting and coefficient extraction
censordata	censordata.R	89	Simulate dropout with outcome-dependent censoring
cumulative	utilities.R	105	Convert intervals to cumulative time (shared)
modgompertz	utilities.R		Modified Gompertz growth curve
reknitsimresults	utilities.R		Recombine chunked simulation outputs
PlotModelingResults	plottingfunctions.R	341	Heatmap visualization of power results
plotfactortrajectories	plottingfunctions.R		Mean trajectory plots by factor

Non-exported / legacy files:

File	Lines	Purpose
comb.R	1,571	Combined scratch: duplicated <code>lme_analysis</code> , <code>generateData</code> , assorted utilities, dead code
scratch_WorkingScript.R	142	Development notes and parameter exploration
datadocumentation.R	61	roxygen2 stubs for embedded datasets
packagedocumentation.R	19	Package-level documentation

1.3.2 main – Key Functions (in `pm_functions.R`)

Function	Purpose
<code>mod_gompertz</code>	Modified Gompertz (reformulated)
<code>calculate_carryover</code>	Multi-model decay (exp, linear, Weibull)
<code>calculate_bio_response_with_interaction</code>	BR means with biomarker interaction
<code>calculate_component_halfives</code>	Component-specific half-life mapping
<code>apply_carryover_to_component</code>	Apply decay to BR component
<code>build_correlation_matrix</code>	Full correlation structure with BM interaction
<code>generate_data</code>	Primary MVN data generation
<code>build_sigma_matrix</code>	Standalone sigma builder for caching
<code>validate_correlation_structure</code>	Eigenvalue diagnostics for sigma
<code>validate_parameter_grid</code>	Pre-simulation PD validation
<code>lme_analysis</code>	Mixed-effects model fitting

1.3.3 simple – Functions (in `simulation.R`)

Function	Purpose
<code>create_design</code>	Trial design specification (switch-based)
<code>randomize_assignment</code>	Per-participant arm randomization
<code>calc_carryover</code>	Simple exponential decay
<code>generate_response</code>	Single-timepoint signal computation
<code>generate_participant</code>	Full participant trajectory
<code>generate_trial</code>	N-participant trial generation
<code>analyze_trial</code>	Design-specific ANCOVA analysis

1.4 Dependency Comparison

Package	orig	main	simple	Notes
data.table	D			Core data structure (orig)
lme4	D	I		<code>lmer()</code> fitting
lmerTest	D	I		p-values for <code>lmer</code>
corpcor	D	I		PD checking/enforcement
ggplot2	D	I		Visualization
MASS	D	I		<code>mvrnorm()</code> for MVN draws
svMisc	D			Progress bar (legacy)

Package	orig	main	simple	Notes
tictoc	D			Timing utilities
ggpubr	D			Publication-quality plots
gridExtra	D			Multi-panel layouts
nlme		I		<code>lme()</code> with <code>corCAR1</code>
dplyr		I	*	Data manipulation
tidyr		*	*	(via tidyverse)
readr		I	*	CSV I/O
tidyverse		I	*	Meta-package (simple uses)
furrr / future		I		Parallel iteration
foreach		I		Parallel loops (alternative)
doParallel		I		Parallel backend
zoo		I		<code>na.locf</code> , time-series
patchwork		I		Plot composition
viridis		I		Color scales
scales		I		Axis formatting
here		I	*	Working directory management
renv		I	I	Package management
testthat		I	I	Testing framework
knitr	S	I		Document rendering
rmarkdown	S	I		Rmarkdown support
bookdown		I		Long-form documents
DBI/odbc/etc.		I		Database (unused in sims)
palmerpenguins		I		Example data (template)
skimr		I		Data summaries
naniar/visdat		I		Missing data visualization
emmeans		I		Estimated marginal means
jsonlite		I		JSON processing
tibble		I	*	(via tidyverse)

Key: D = Depends, I = Imports, S = Suggests, * = implicit via tidyverse

1.5 Test Inventory

Repository	Test File	Coverage	Assertions
orig	(none)	–	0
main	test-utils.R	Trivial pass-through	1
main	test-analysis-scripts.R	Script executability	~3
main	test-data-pipeline.R	Data workflow pipeline	~3
main	test-report-rendering.R	Rmd rendering	~2
simple	test-basic.R	Trivial pass-through	1

All three repositories have effectively zero unit-test coverage of the core simulation logic (DGP, carryover, correlation matrix, analysis model). The main repository has integration tests that verify scripts execute without error, but do not validate statistical correctness.

2. Statistical Model Comparison

2.1 Response Decomposition

All three codebases decompose the outcome as:

$$Y_i(t) = BL_i - [BR_i(t) + ER_i(t) + TR_i(t)]$$

where BL is baseline symptom severity and the bracketed terms represent improvement (higher = more improvement = lower symptoms). The sign convention treats the outcome as a symptom score that decreases with effective treatment.

2.1.1 Biological Response (BR)

Aspect	orig	main	simple
Growth curve	Modified Gompertz	Modified Gompertz (reformulated)	Linear rate
Formula (on-drug)	<code>modgompertz(tod, max, disp, rate)</code>	<code>mod_gompertz(tod, max, disp, rate)</code>	<code>br_rate * (1 + bm_mod * bm)</code>
Gompertz form	<code>max*exp(-d*exp(-r*t)) - offset</code>	<code>max * (1 - exp(-d*exp(-r*t)))</code>	N/A (linear)
Biomarker mod.	Via correlation (c.bm in sigma)	Via correlation (c.bm in sigma)	Via mean structure (explicit term)
Time dependence	Cumulative time-on-drug	Cumulative time-on-drug	Per-week (not cumulative)

Critical difference in Gompertz formulation. The orig uses a traditional Gompertz with a vertical offset subtraction and rescaling:

```
# orig: modgompertz
y <- maxr * exp(-disp * exp(-rate * t))
vert_offset <- maxr * exp(-disp * exp(-rate * 0))
y <- y - vert_offset
y <- y * (maxr / (maxr - vert_offset))
```

The main reformulates this as:

```
# main: mod_gompertz
y <- max_value * (1 - exp(-displacement * exp(-rate * time)))
```

These two formulations are NOT algebraically identical. The orig produces a curve that passes through zero at $t=0$ and approaches `maxr` asymptotically, but with different curvature than the main's version. The main's formulation is a complementary Gompertz that also passes through zero at $t=0$ but with a different displacement parameterization. Whether the `disp` and `rate` parameter values are recalibrated to produce equivalent trajectories is not documented.

The simple version abandons the Gompertz entirely and uses a constant rate model where BR is proportional to `br_rate` with no saturation, yielding a fundamentally different dose-response shape.

2.1.2 Expectancy Response (ER)

Aspect	orig	main	simple
Growth curve	Modified Gompertz	Modified Gompertz (reformulated)	Constant rate
Input variable	<code>tpb</code> (time in belief period)	<code>tpb</code> (time in belief period)	Binary (unblinded + on-drug)
Expectancy wt.	Scaled by <code>e</code> (0-1 probability)	Scaled by expectancy parameter	Binary: <code>er_rate</code> or 0
SD scaling	<code>sd * e</code> (reduces variance too)	Similar expectancy scaling	Fixed <code>sigma_within</code>

2.1.3 Time-Variant Response (TR)

Aspect	orig	main	simple
Growth curve	Modified Gompertz	Modified Gompertz (reformulated)	Absent
Input variable	<code>t_wk_cumulative</code> (absolute time)	Cumulative elapsed time	–
Purpose	Natural disease progression / drift	Natural disease progression / drift	Assumed stable baseline

The simple version omits TR entirely, assuming no natural progression over the trial duration. This is a meaningful simplification that removes one source of confounding from the simulation.

2.2 Covariance Structure

Aspect	orig	main	simple
Matrix dimension	$(2 + 3nTP) \times (2 + 3nTP)$ e.g., 26x26 for 8 timepoints	$(2 + 3nTP) \times (2 + 3nTP)$ e.g., 26x26 for 8 timepoints	N/A
Autocorrelation (BR)	Constant: <code>c.br</code>	Time-based AR(1): <code>c.br^abs(t_i - t_j)</code>	None (iid)
Autocorrelation (ER)	Constant: <code>c.pb</code>	Time-based AR(1): <code>c.pb^abs(t_i - t_j)</code>	None (iid)
Autocorrelation (TR)	Constant: <code>c.tv</code>	Time-based AR(1): <code>c.tv^abs(t_i - t_j)</code>	None
Cross-factor (same t)	<code>c.cf1t</code>	<code>c.cf1t</code>	None
Cross-factor (diff t)	<code>c.cfct</code>	<code>c.cfct * 0.9^time_lag</code>	None
BM-BR correlation	<code>c.bm</code> (with Ron Thomas adj.)	<code>c.bm</code> (on-drug only, 0 off-drug)	N/A (mean structure)

Aspect	orig	main	simple
PD enforcement	<code>corpcor::make.positive.definite</code>	<code>corpcor::make.positive.definite</code>	Not needed
PD tolerance	<code>tol = 1e-3</code>	<code>tol = 1e-3</code>	—
Data generation	<code>MASS::mvrnorm</code>	<code>MASS::mvrnorm</code>	<code>rnorm</code> (indep. components)
Variance model	Full MVN (all components jointly)	Full MVN (all components jointly)	Random intercept + iid

Key structural differences:

1. The orig uses **constant** autocorrelation across all time lags (compound symmetry within each factor). The main uses **time-based AR(1)** where correlation decays with actual elapsed time. This is a substantive methodological improvement in the main.
2. The orig's biomarker-BR correlation uses a Ron Thomas modification that scales `c.bm` by a ratio of means (`mm1/mm0`) when `brtest[p]` is TRUE (i.e., when the raw BR mean is zero at a timepoint). The main simplifies this to a binary rule: `c.bm` when on treatment, 0 when off treatment.
3. The simple bypasses the covariance structure entirely, placing the biomarker interaction directly in the mean structure (`br_rate * (1 + bm_mod * bm)`) rather than encoding it as a correlation. This is a fundamentally different statistical mechanism for the biomarker-treatment interaction.

2.3 Carryover Model

Aspect	orig	main	simple
Decay function	Exponential only	Exponential, Linear, Weibull	Exponential only
Formula (exponential)	<code>BR[p-1] * (0.5)^(sf * tsd / t_half)</code>	<code>prev_effect * (0.5)^(sf * tsd / t_half)</code>	<code>0.5^(tsd / t_half)</code>
Scale factor default	2	2	1 (implicit)
Half-life values	Not specified in code (set externally)	0, 0.1, 0.2 weeks	0, 0.5, 1, 2 weeks
Components affected	BR only	BR only	BR only
Boundary handling	Skipped if <code>tsd == 0</code>	Skipped if off-drug time is zero	Returns 0 if <code>halflife <= 0</code>
Where applied	DGP means (<code>generateData</code>)	DGP means (<code>pm_functions</code>) AND analysis model	DGP (<code>generate_response</code>) AND analysis

Scale factor discrepancy. The orig and main use `scalefactor = 2` by default, which doubles the effective decay rate. At `t_half = 0.1` weeks with `scalefactor = 2`, the effective half-life is

0.05 weeks (~8.4 hours). The simple omits the scale factor (effectively `scalefactor = 1`), so its decay at `t_half = 0.5` weeks gives a true half-life of 0.5 weeks. These are not comparable without adjustment.

Half-life value divergence. The main’s CLAUDE.md explicitly warns against using the values 0.5, 1.0, 2.0 weeks, calling them “legacy” and “incorrect.” Yet the simple uses exactly these values. The simple’s CLAUDE.md inherits this warning but the code contradicts it.

2.4 Trial Designs Supported

Design	orig	main	simple
Open-Label (OL)	Yes	Yes	Yes*
Open-Label + Blinded Disc.	Yes	Yes	Yes
Traditional Crossover (CO)	Yes	Yes	Yes
Hybrid (N-of-1)	Yes	Yes	Yes
Parallel	Yes	Yes	No

*OL is defined in simple but excluded from the main simulation because it has no drug comparison arm.

Design specification mechanism:

- orig: User-specified via `buildtrialdesign()` with arbitrary timepoints, expectancy vectors, and ondrug path lists. Highly flexible.
- main: Hard-coded design functions in simulation scripts (e.g., `create_hybrid_design()`). Less flexible but more reproducible for the specific comparison.
- simple: Switch-based `create_design()` with four pre-defined designs. Simplest but least extensible.

2.5 Analysis Model

Aspect	orig	main	simple
Fitting function	<code>lme4::lmer()</code>	<code>lme4::lmer()</code> or <code>nlme::lme()</code>	<code>stats::lm()</code>
Correlation structure	None (random effects only)	Optional <code>corCAR1</code> (nlme only)	None
Random effects	<code>(1 ptID)</code> or <code>(1+t ptID)</code>	<code>(1 participant_id)</code>	None (ANCOVA on means)
Fixed effects (CO/HY)	<code>Sx ~ bm + t + Dbc + bm*Dbc</code>	<code>response ~ eff_drug_wks * bm_centered + week</code>	<code>drug_effect ~ biomarker</code>
Fixed effects (OL)	<code>Sx ~ bm + t + bm*t</code>	<code>response ~ week * bm_centered</code>	<code>mean_response ~ biomarker</code>
Interaction term	<code>bm:Dbc</code> or <code>bm:t</code>	<code>effective_drug_weeks:bm_centered</code>	<code>biomarker (main effect on diff)</code>
Carryover in analysis	<code>Dbc = (0.5)^(sf*tsd/t_half)</code> (continuous)	<code>effective_drug_weeks</code> (decayed cumulative)	Optional subtraction from response

Aspect	orig	main	simple
Two-scenario design	Single model per condition	Yes: <code>model_carryover</code> TRUE vs FALSE	Yes: <code>carryover_adjust</code> TRUE vs FALSE

Key analytical differences:

1. The orig constructs a continuous drug indicator `Dbc` that equals 1 on-drug and decays exponentially off-drug. The biomarker interaction is then `bm * Dbc`. This is an elegant approach that embeds carryover directly into the predictor variable.
2. The main constructs `effective_drug_weeks` (cumulative drug exposure with decay) and tests `effective_drug_weeks * bm_centered`. It also compares this against a misspecified model using `weeks_on_drug` (no decay) to quantify the cost of ignoring carryover.
3. The simple collapses repeated measures to phase means and runs simple linear regression. This sacrifices within-subject temporal information but eliminates all convergence issues associated with mixed-effects models.

2.6 Censoring / Dropout

Aspect	orig	main	simple
Implemented	Yes (<code>censordata()</code>)	Yes (<code>simulation_censoring.R</code>)	Yes (per-week probability)
Mechanism	Time-dependent + outcome-dependent	Multiple patterns	MCAR only
Parameters	<code>beta0</code> (rate), <code>beta1</code> (outcome fraction), <code>eb1</code> (exponent)	5 censoring scenarios	<code>dropout_prob</code> (0-0.10/week)
Time dependence	Proportional to interval duration	Various	Constant per week
Outcome dependence	Yes (weighted by delta scores)	Yes (some scenarios)	No (MCAR)
Implementation	Post-hoc NA insertion	Post-hoc NA insertion	Row filtering

2.7 Side-by-Side Summary

Component	orig	main	simple
Growth curve	Modified Gompertz (offset-rescale)	Modified Gompertz (complementary)	Linear rate
Factors modeled	BR, ER, TR	BR, ER, TR	BR, ER
BM interaction	Correlation-based	Correlation-based	Mean-based

Covariance	Compound symmetry	AR(1), time-based	Indep. + RE	
+-----+	+-----+	+-----+	+-----+	+-----+
Carryover models	Exponential	Exp, Linear, Weibull	Exponential	
+-----+	+-----+	+-----+	+-----+	+-----+
Scale factor	2 (default)	2 (default)	1 (implicit)	
+-----+	+-----+	+-----+	+-----+	+-----+
Half-lives (wk)	External	0, 0.1, 0.2	0, 0.5, 1, 2	
+-----+	+-----+	+-----+	+-----+	+-----+
Analysis model	lmer	lmer or lme+corCAR1	lm (ANCOVA)	
+-----+	+-----+	+-----+	+-----+	+-----+
Censoring	Time + outcome	Multiple patterns	MCAR only	
+-----+	+-----+	+-----+	+-----+	+-----+
Parallelism	Sequential	furrr/future	Sequential	
+-----+	+-----+	+-----+	+-----+	+-----+
PD enforcement	corpcor	corpcor	Not needed	
+-----+	+-----+	+-----+	+-----+	+-----+

3. Code Quality and Architecture Audit

3.1 Design Patterns

3.1a orig

- **Modularity:** Moderate. Functions are separated into logical files (design, data gen, analysis, plotting), but `comb.R` (1,571 lines) duplicates much of the other code and contains scratch work.
- **Parameter specification:** Data.table-based parameter structures with named columns. No formal validation.
- **Parallelism:** Sequential only. The `svMisc` progress bar and `tictoc` timing are the only concessions to long runtimes.
- **Output management:** Progressive RDS saving with numbered suffixes (`_save1`, `_save2`, etc.). Fragile naming convention.
- **Reproducibility:** No Docker, `renv`, or Makefile. Depends on the user having all packages installed at compatible versions.

3.1b main

- **Modularity:** Good separation between shared functions (`pm_functions.R`) and simulation scripts. However, the simulation scripts are large monoliths (1,233 lines for `simulation_clustered.R`) that inline much logic that could be factored out.
- **Parameter specification:** Explicit parameter grids with `expand.grid()` or `expand_grid()`. Pre-simulation validation via `validate_parameter_grid()`.
- **Parallelism:** `furrr::future_map_dfr()` for iteration-level parallelism. Configurable via `future::plan()`.
- **Output management:** Timestamped filenames, CSV export, structured logging to file.
- **Reproducibility:** Docker + `renv` + Makefile. The most robust of the three.

3.1c simple

- **Modularity:** Single file by design. All functions, parameters, and execution in one script. Readable but not reusable.
- **Parameter specification:** Hard-coded constants at script top. Clean and transparent.
- **Parallelism:** None. Acceptable given ~4 minute runtime.
- **Output management:** Timestamped CSV and PNG output.
- **Reproducibility:** Docker + renv + Makefile + CI/CD via GitHub Actions.

3.2 Code Issues

3.2a orig – Issues

1. **Dead code: `comb.R` (1,571 lines).** This file contains duplicated versions of `lme_analysis()`, `generateData()`, and various utilities alongside scratch work and commented-out experiments. It is actively loaded by the package (listed in `R/`) and creates namespace pollution and confusion about which version of a function is authoritative.
2. **`eval(parse(...))` in `generateData.R` (lines 204-214).** The outcome computation uses string construction and `eval(parse())` to dynamically build `data.table` expressions:

```
evalstring <- paste(
  "dat[,D_", n, "]:=sum(",
  paste(paste(n, cl, sep='.', collapse=''),
  ")", by='ptID']", sep="")
eval(parse(text=evalstring))
```

This is fragile, hard to debug, and a potential injection vector if inputs are not sanitized. It should be replaced with programmatic `data.table` column operations.

3. **Hard-coded indices in `generateData.R` (line 89).** `names(brtest) <- labels[19:26]` assumes exactly 8 timepoints and 2 baseline variables. This breaks for any other trial design length.
4. **Missing error handling.** No `tryCatch()` around `lmer()` calls (the code comments note “should insert `tryCatch` here, but not working!”). No validation of input parameters. No check for degenerate covariance matrices before `mvnrm()`.
5. **`scratch_WorkingScript.R` in `R/`.** Development notes do not belong in the package source directory.
6. **Unqualified function calls.** Functions from `data.table`, `MASS`, `lme4`, and `corpcor` are called without namespace qualification, relying on `Depends:` to attach them. This is fragile and against modern R packaging conventions.
7. **No tests whatsoever.** Zero test files.
8. **Symmetry bug (fixed in-place).** Line 134 of `generateData.R` contains a note about fixing a typo where both `correlations[n1,n2]` and `correlations[n2,n1]` were set with the same indices, breaking matrix symmetry.

3.2b main – Issues

1. **Massive DESCRIPTION dependency list.** 38 packages in Imports, including database drivers (`DBI`, `odbc`, `RMySQL`, `RPostgres`, `RSQLite`), example datasets (`palmerpenguins`), and tools

(`conflicted`, `sessioninfo`, `skimr`, `naniar`, `visdat`) that are never used in the simulation code. This is a `zzcollab` template artifact that inflates the dependency footprint.

2. **Empty NAMESPACE.** The package exports nothing. All functional code lives in `analysis/scripts/` rather than `R/`. This means the “package” structure is vestigial; it functions as a research compendium, not an installable R package.
3. **Proliferation of simulation script variants.** There are 13+ simulation scripts (`simulation_clustered.R`, `simulation_clustered_exponential.R`, `simulation_clustered_flexible_correlation.R`, `simulation_clustered_twoweek.R`, `simulation_evenly_spaced.R`, `simulation_evenly_spaced_flexible_correlation.R`, `simulation_evenly_spaced_powerlaw.R`, `simulationplus.R`, etc.) with substantial code duplication. Shared logic (design creation, parameter grid setup, output formatting) is copied across files rather than factored into `pm_functions.R`.
4. **Ad-hoc test scripts in analysis/scripts/.** Files like `test_autocorr_for_cbm_06.R`, `test_boundary_autocorr_06.R`, `test_cbm_06.R`, `test_new_params.R`, `test_pd_boundaries.R` are not integrated into the `testthat` framework. They are standalone exploration scripts that could silently break.

5. **Trivial test coverage.** The single `testthat` test is:

```
test_that("package loads correctly", {  
  expect_true(TRUE)  
})
```

This provides zero coverage of the 1,301-line `pm_functions.R` or any simulation logic.

6. **Documentation volume vs. coherence.** Over 40 documentation files exist, but many overlap, some contradict each other, and there is no clear canonical reference. A new collaborator would struggle to identify which document is authoritative.
7. **Deprecated function still present.** `calculate_carryover_adjusted_correlations()` is marked deprecated in comments but may still be called from older simulation scripts.

3.2c simple – Issues

1. **Carryover half-life values contradict project convention.** The `CLAUDE.md` (inherited from main) states “Do NOT use the legacy values (0.5, 1.0, 2.0 weeks)” but `simulation.R` line 37 uses exactly `c(0, 0.5, 1, 2)`.
2. **Missing DESCRIPTION dependency.** The script requires `tidyverse` via `library(tidyverse)` but `DESCRIPTION` lists only `renv` and `testthat` as imports. This would fail R CMD check.
3. **Trivial test coverage.** Same `expect_true(TRUE)` placeholder as main.
4. **Placeholder author in DESCRIPTION.** The author field contains “Your Name” and “your.email@example.com” from the `zzcollab` template.
5. **here::here() used without here in Imports.** Works in the Docker container (via `tidyverse` transitive dependency) but is not formally declared.
6. **Magic number thresholds.** Minimum sample sizes for analysis (10 for OL+BDC, 5 for CO/N-of-1) are hard-coded without justification.

3.3 Documentation Quality

3.3a orig

- **Completeness:** The vignettes describe the workflow (generate, visualize, apply to real data) but do not fully specify the DGP mathematics. The modified Gompertz, carryover formula, and correlation structure are documented only in code comments and roxygen headers.
- **Accuracy:** The code is the ground truth. The `scalefactor` parameter is documented as “TODO update when understand what this does?” (line 43 of `generateData.R`), indicating incomplete understanding even by the original author.
- **Accessibility:** A new collaborator would need to read the code to understand the simulation. The vignettes provide usage examples but not mathematical derivations.

3.3b main

- **Completeness:** Extensive. Over 40 documents cover the mathematical foundations, carry-over theory, sigma matrix derivation, positive definiteness constraints, Hendrickson alignment, and mixed model comparison. LaTeX documents provide formal mathematical treatment.
- **Accuracy:** Generally good, but the volume creates risk of contradictions. The critical finding about carryover half-life scale is well-documented in `CRITICAL_FINDING_CARRYOVER_HALFLIFE.md`.
- **Accessibility:** Mixed. The sheer volume (40+ files) is overwhelming. There is no single entry-point document that provides a complete mathematical specification. `docs/readme.md` exists but may not index all documents. A new collaborator could reproduce the simulation from documentation, but would need significant time to navigate the corpus.

3.3c simple

- **Completeness:** The `simplification-plan.md` document thoroughly explains what was simplified and why. The `comparison_whitepaper.md` compares all three codebases. However, the DGP is documented only in code comments.
- **Accuracy:** The code is straightforward enough that the documentation matches. The carryover half-life mismatch with `CLAUDE.md` is the notable exception.
- **Accessibility:** Good. A single 562-line script with inline documentation is easy to follow. A new collaborator could understand the simulation quickly.

4. Evolution Analysis

4.1 Changes from orig to main

Category	Change	Significance
Architecture	Package functions to research compendium scripts	Major
Data structures	<code>data.table</code> to <code>tibble/tidyverse</code>	Moderate
Gompertz formula	Offset-rescale to complementary form	Moderate
Autocorrelation	Compound symmetry to time-based AR(1)	Major
Cross-correlation	Constant to time-decaying (0.9^{lag})	Moderate
BM correlation	Ron Thomas mean-ratio adj. to binary on/off	Major
Carryover models	Exponential only to Exp + Linear + Weibull	Moderate
Carryover values	Unspecified to 0, 0.1, 0.2 weeks (corrected)	Critical

Category	Change	Significance
Analysis engine	lmer only to lmer + lme with corCAR1	Major
Two-scenario	Single model to WITH/WITHOUT carryover comparison	Major
Parallelism	Sequential to furrr/future	Moderate
Sigma validation	None to pre-simulation eigenvalue diagnostics	Major
Sigma caching	Rebuilt per iteration to pre-computed cache	Moderate
Dependencies	10 packages to 38+ packages	Negative
Dead code	comb.R present to (mostly) removed	Positive
eval(parse)	Present to eliminated	Positive
Docker/renv	Absent to present	Major
Documentation	4 files to 40+ files	Major

4.2 Simplifications from main to simple

Simplification	Statistical validity preserved?
MVN covariance to random intercept	Partially. Loses temporal correlation structure; overstates effective N and thus overstates power.
Modified Gompertz to linear rate	Partially. Valid for short trials where saturation is negligible; misrepresents dose-response shape for longer trials.
Correlation-based BM interaction to mean-based interaction	Different mechanism. The mean-based approach is more transparent but produces a different effect-size interpretation than the MVN approach.
lmer/lme to lm on phase means	Partially. Loss of temporal resolution reduces power but avoids convergence issues. The estimand changes from per-timepoint to phase-mean difference.
TR removed	Valid if natural progression is negligible over trial duration (~20 wk).
Outcome-dependent dropout to MCAR	Partially. MCAR preserves unbiasedness but misses the bias introduced by informative dropout.
Carryover half-lives 0.1/0.2 to 0.5/1/2 weeks	Invalid for comparison. The values are 5-20x longer, producing much slower decay and larger carryover effects than Hendrickson.
Scale factor 2 to 1 (implicit)	Changes the effective decay rate. Combined with longer half-lives, this produces a categorically different carryover regime.

4.3 Backport Candidates

Improvement	From	To	Priority
Pre-simulation sigma validation	main	orig	High
Time-based AR(1) autocorrelation	main	orig	High
<code>eval(parse())</code> elimination	main	orig	High
Docker + renv infrastructure	main	orig	Medium
Carryover half-life correction (0.1/2)	main	simple	Critical
CI/CD pipeline (GitHub Actions)	simple	main	Medium
Single-file pedagogical script	simple	main	Low
Weibull/linear carryover options	main	simple	Low

4.4 Unresolved Divergences

4.4.1 Gompertz formulation The orig and main use different mathematical forms of the modified Gompertz function with the same parameter names (`max`, `disp/displacement`, `rate`). Unless the parameter values are recalibrated, the two produce different response trajectories for the same inputs. **The main’s formulation appears correct** as a standard complementary Gompertz, while the orig’s offset-rescale approach is non-standard and harder to interpret.

4.4.2 Biomarker-treatment interaction mechanism The orig and main encode the interaction via differential correlation in the MVN structure (`c.bm > 0` on-drug, reduced/zero off-drug). The simple encodes it as an additive term in the mean structure (`br_rate * bm_mod * bm`). These produce different estimands: the MVN approach creates a correlation-mediated association that manifests as a regression coefficient, while the mean-based approach creates a direct moderation effect. **Both are defensible** but they answer slightly different statistical questions and produce non-comparable power estimates.

4.4.3 Carryover half-life values

- main: 0, 0.1, 0.2 weeks (with scale factor 2)
- simple: 0, 0.5, 1, 2 weeks (with implicit scale factor 1)

Effective half-lives: - main: 0.05, 0.10 weeks (~8h, ~17h) - simple: 0.5, 1.0, 2.0 weeks (~84h, ~168h, ~336h)

The simple’s carryover persists 10-20x longer than the main’s. **The main’s values match Hendrickson et al. (2020) and are correct for the intended application.** The simple’s values may be deliberately chosen to demonstrate carryover effects more visibly, but they are not comparable to the published methodology.

4.4.4 Autocorrelation structure

- orig: Compound symmetry (constant correlation across lags)
- main: AR(1) with time-based decay
- simple: Independence (no temporal correlation)

The main’s AR(1) structure is most realistic for longitudinal biomarker data. The orig’s compound symmetry is an overly strong assumption. The simple’s independence assumption is appropriate for its pedagogical purpose but would overstate power in realistic applications.

5. Recommendations

5.1 Statistical Correctness (Priority: Critical)

1. **Harmonize carryover half-life values in simple.** Replace `c(0, 0.5, 1, 2)` with `c(0, 0.1, 0.2)` to match Hendrickson et al. (2020) and the main codebase. If the longer values are retained for pedagogical purposes, they must be clearly labeled as non-Hendrickson exploratory values, and the scale factor difference must be documented.
2. **Verify Gompertz equivalence between orig and main.** Run both formulations with identical parameters and compare response trajectories. If they diverge, document which is authoritative and recalibrate parameters in the non-canonical version.
3. **Document the biomarker interaction mechanism difference.** The correlation-based (orig, main) and mean-based (simple) approaches are not equivalent. A brief technical note comparing the two mechanisms and their implications for power estimation would prevent inappropriate cross-codebase comparisons.
4. **Validate type I error rates.** None of the codebases includes a null-case validation (biomarker moderation = 0, expected power = alpha). This is essential for confirming that the simulation and analysis pipeline does not produce inflated false-positive rates.

5.2 Software Engineering (Priority: High)

5. **Add unit tests for core simulation functions.** At minimum, test:
 - Gompertz function: known input/output pairs
 - Carryover decay: boundary conditions ($t=0$, $t=\infty$, $halflife=0$)
 - Correlation matrix: symmetry, PD, correct dimensions
 - Analysis model: coefficient recovery under known effect sizes
 - Null case: type I error rate within binomial bounds
6. **Eliminate `comb.R` from orig.** This 1,571-line file is dead weight. Archive it outside the R/ directory or delete it entirely.
7. **Eliminate `eval(parse())` from orig.** Replace with programmatic `data.table` column operations using `.SDcols` or `set()`.
8. **Reduce main's DESCRIPTION dependencies.** Remove database drivers, `palmerpenguins`, `skimr`, `naniar`, `visdat`, and other packages not used in any simulation or analysis script.
9. **Consolidate simulation script variants in main.** Factor shared logic (design creation, parameter grid setup, model fitting, output formatting) into `pm_functions.R` and reduce the 13+ scripts to 2-3 canonical entry points with configuration parameters.
10. **Fix simple's DESCRIPTION.** Add `tidyverse` (or specific tidyverse packages) to Imports. Replace placeholder author information.

5.3 Documentation (Priority: Medium)

11. **Create a canonical mathematical specification.** A single LaTeX document (or structured Rmd) that fully specifies the DGP, analysis model, and power calculation. This should be the authoritative reference that all three codebases implement, with a correspondence table mapping mathematical notation to variable names in each codebase.

12. **Create a documentation index for main.** The 40+ documents need a clear hierarchy: which is authoritative, which is historical, which is exploratory. A single `docs/INDEX.md` with status annotations would suffice.
13. **Document the scale factor.** The orig’s roxygen for `scalefactor` says “TODO update when understand what this does?” This parameter fundamentally changes the carryover regime and must be properly documented in all three codebases.

5.4 Consolidation Strategy (Priority: Strategic)

The three repositories serve distinct purposes and should be maintained as separate artifacts, but with tighter alignment:

- **orig:** Archive. Freeze as the published reference implementation for Hendrickson et al. (2020). Do not develop further. Apply only the minimal fixes needed for reproducibility (renv, Docker, elimination of `comb.R`).
- **main:** Primary research codebase. Continue active development. Consolidate simulation variants, expand test coverage, and produce a canonical mathematical specification document. This is the codebase that should be used for publication-quality results.
- **simple:** Pedagogical companion. Align carryover parameters with Hendrickson. Add a comparison mode that runs both mean-based and correlation-based biomarker interaction mechanisms side by side. This codebase should prioritize clarity and be usable as teaching material or a supplementary methods appendix.

A shared `pmsim-common` package containing only the mathematical primitives (Gompertz, carryover decay, correlation matrix builder) would reduce drift across the three codebases, but the overhead of maintaining a fourth repository may not be justified given the current team size.

Appendix A: Function Signature Cross-Reference

orig	main	simple
<code>modgompertz(t, maxr, disp, rate)</code>	<code>mod_gompertz(time, max_value, disp, rate)</code>	<code>(linear: br_rate * (1 + bm_mod * bm))</code>
<code>(inline in generateData)</code>	<code>calculate_carryover(tsd, prev_effect, model, params)</code>	<code>calc_carryover(weeks_since_drug, halflife)</code>
<code>generateData(modelparam, respparam, blparam, trialdesign, empirical, makePositiveDefinite, seed, scalefactor)</code>	<code>generate_data(model_param, resp_param, baseline_param, trial_design, empirical, make_pd, seed, scale_factor)</code>	<code>generate_participant(id, design, biomarker, ...)</code>

buildtrialdesign(name,	create_*_design(	create_design(	
timepoints, expectancy,	n_participants,	design_name)	
ondrug)	measurement_weeks)		
+-----+	+-----+	+-----+	+-----+
lme_analysis(td_set,	lme_analysis(td_set,	analyze_trial(	
dat, op)	data, options)	trial_data,	
		design_name,	
		adjust_carryover)	
+-----+	+-----+	+-----+	+-----+
censordata(dat, td,	(in simulation scripts)	(inline in	
censorparam)		generate_participant	
+-----+	+-----+	+-----+	+-----+
generateSimulatedResults(	(in simulation scripts)	(inline main loop)	
trialdesigns, resp,			
bl, censor, model,			
sim, analysis)			
+-----+	+-----+	+-----+	+-----+

Appendix B: Parameter Defaults

+-----+	+-----+	+-----+	+-----+
Parameter	orig	main	simple
+-----+	+-----+	+-----+	+-----+
N (participants)	external	35, 70	35, 70
Carryover half-life (wk)	external	0/0.1/0.2	0/.5/1/2
Scale factor	2	2	1
c.bm (BM correlation)	external	0.3	N/A
c.br (BR autocorrelation)	external	0.75	N/A
c.pb (ER autocorrelation)	external	0.75	N/A
c.tv (TR autocorrelation)	external	0.75	N/A
c.cflt (cross-factor)	external	0.12	N/A
c.cfct (cross-time)	external	0.05	N/A
sigma_between	external	(in MVN)	1.0
sigma_within	external	(in MVN)	0.5
Iterations	external	500	500
alpha	0.05	0.05	0.05
Designs	flexible	4 fixed	3 fixed
+-----+	+-----+	+-----+	+-----+

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